

BDF - Customized Product Documentation

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BDF Product Documentation: Working with Hermary Amadeus Cameras for Bakery Automation

Welcome to the BDF product documentation for Hermary Amadeus Cameras. These advanced camera systems are designed to integrate seamlessly into your bakery automation processes, providing robust tools for quality control, detailed inspection, and precise product monitoring.

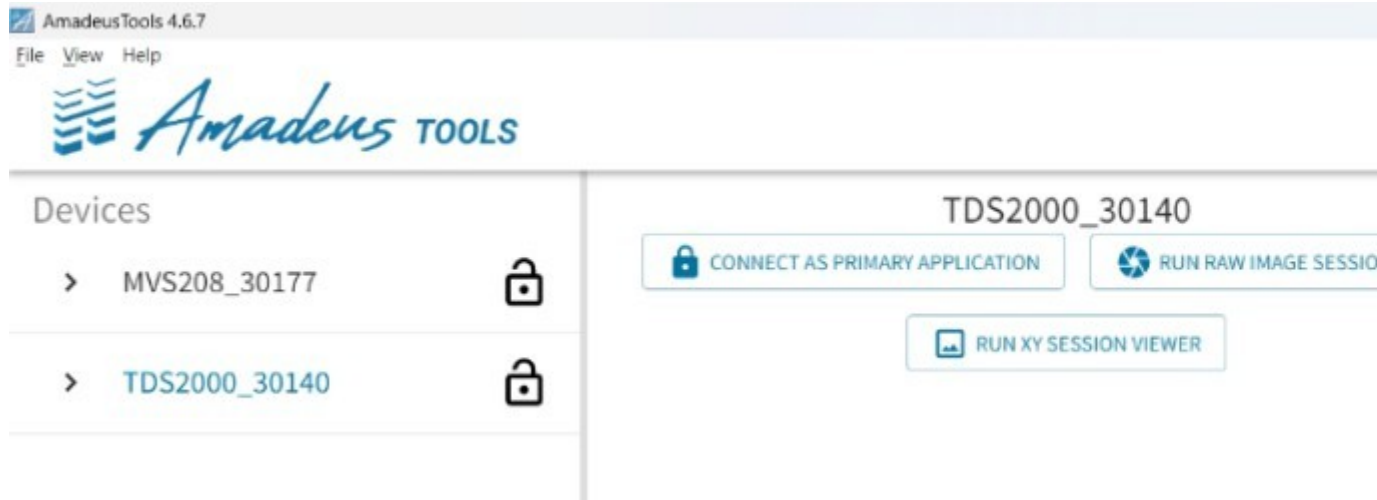
This document will guide BDF users through the two main software components: **Amadeus Tools** and **The Composer**, ensuring you can efficiently set up, manage, and optimize your camera systems for maximum production efficiency and product quality.

Amadeus Tools: Your Camera Management Hub

Amadeus Tools is the primary interface for managing and monitoring your Hermary Amadeus cameras. It allows BDF operators to connect devices, adjust hardware and software settings, and access live raw imagery from your production lines.

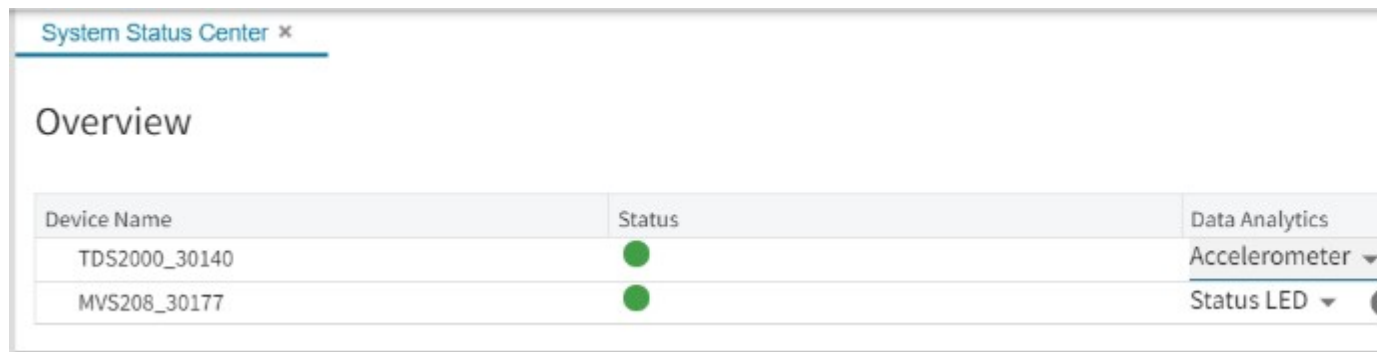
Connecting and Updating Devices

When you first open Amadeus Tools, the system is designed to automatically find any connected devices, such as the MVS208 control module/switch and the TDS2000 3D camera. They will be auto-assigned IP addresses and appear in the "Devices" menu on the left side of the screen, ready for use in your bakery operations.



Checking Device Status

You can quickly verify that your cameras are connected and functioning correctly by checking the "Status" section. Click the button on the right side of the screen to ensure your inspection system is always online and ready for production.



Analyzing Production Data

The "Data Analytics" tab provides valuable insights into your camera's performance. You can select different metrics from the drop-down menu and click the play button to view various charts. This feature helps BDF monitor trends in product consistency, production line efficiency, or other operational parameters.

System Status Center ✕

Overview

Device Name	Status	
TDS2000_30140	●	
MVS208_30177	●	

Logs

Device Name	Date	Severity	Message
No data			

Accelerometer
 PTP
 Status LED
 Uptime
 Temperature
 Voltage

System Status Center ✕

Overview

Device Name	Status	
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No data			

Data Analytics
 Accelerometer
 Status LED
 Uptime
 Temperature
 Voltage

Setting Network Addresses (IPs) Manually

If your BDF network configuration requires specific static IP addresses for your cameras, you can easily set them here.

1. Click on the lock icon next to the device you'd like to modify. The lock should turn green, indicating you are connected to the device.
2. Navigate to "Transport Layer Control" > "GigEVision".
3. Change the Persistent IP Address, Subnet Mask, and Default Gateway as required by your network.

Devices	
> MVS208_30177	
> TDS2000_30140	

MVS208_30177	
SET DEFAULT VALUES	
Search	
> Category	Node Value
> Device Control	
▼ Transport Layer Control	
> PtpControl	
▼ GigEVision	
MAC Address	00:60:ec:00:2a:fe
Current IP set in LLA	True ▼
Current IP set in DHCP	True ▼
Current IP set in Persistent	True ▼
Current IP Address	192.168.137.41
Current IP Subnet Mask	255.255.255.0
Current IP Default Gateway	192.168.137.1
Persistent IP Address	192.168.137.41
Persistent Subnet Mask	255.255.255.0
Persistent Default Gateway	192.168.137.1
Control Channel Privilege	Control Access ▼

>

TDS2000_30140

START CENTROID VIEWER

SET DE

Search

> Category	Node Value
> Device Control	
> Acquisition Control	
> Image Format Control	
▼ Transport Layer Control	
Payload Size	0
> PtpControl	
▼ GigEVision	
Physical Link Configuration Capabili	Dynamic LAG Config
Current Physical Link Configuration	Single Link Configur
MAC Address	00:60:ec:00:2a:b5
Current IP set in LLA	True ▼
Current IP set in DHCP	True ▼
Current IP set in Persistent	True ▼
Current IP Address	192.168.137.42
Current IP Subnet Mask	255.255.255.0
Current IP Default Gateway	192.168.137.1
Persistent IP Address	192.168.137.42
Persistent Subnet Mask	255.255.255.0
Persistent Default Gateway	192.168.137.1

Raw Image Viewer: Live Inspection

After connecting to a TDS2000 device, select the "Start Raw Image Viewer" button to open a live feed of the camera. This is perfect for real-time visual inspection of your bakery products, allowing BDF operators to observe details like dough texture, product shape, or baked crust quality as they move along the production line. You can modify the camera's intensity using the sliders on the bottom dashboard for optimal viewing.

Raw Image Viewer:TDS2000_30140:51200 ×

Timestamp: 1748540355.160000016 s
Encoder Count: 1
Exposure Time: 200 μ s

Intensity

0

25

50

75

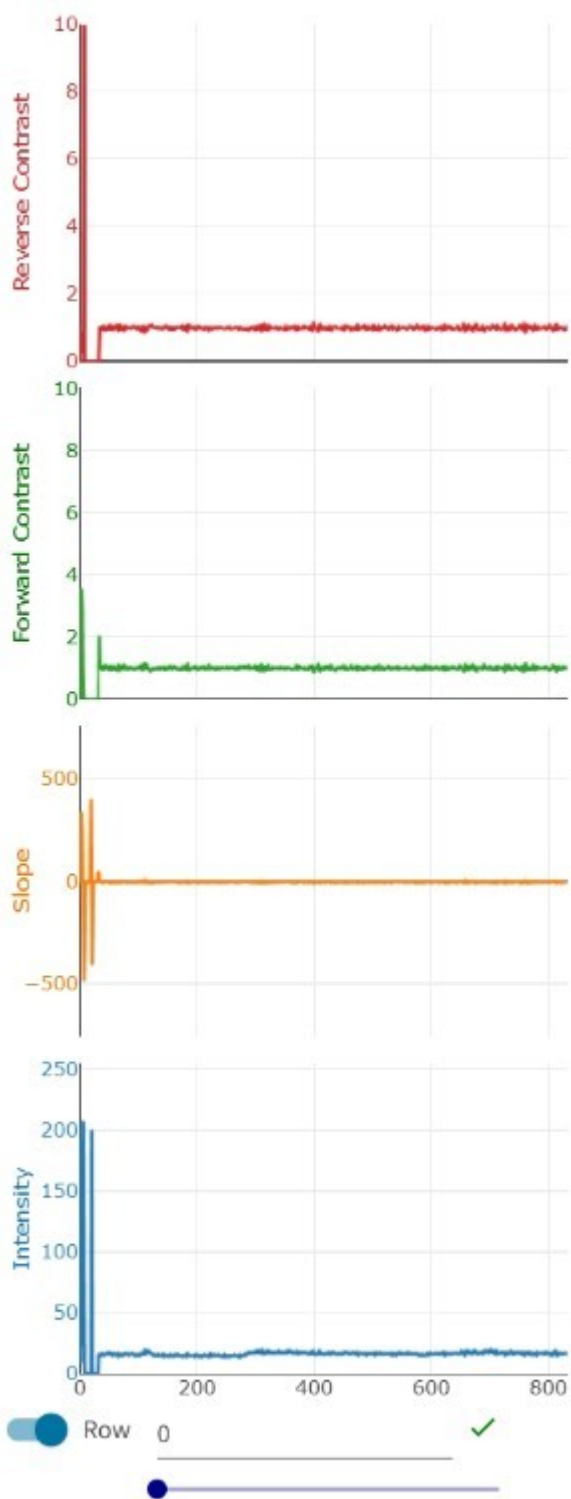
100

0

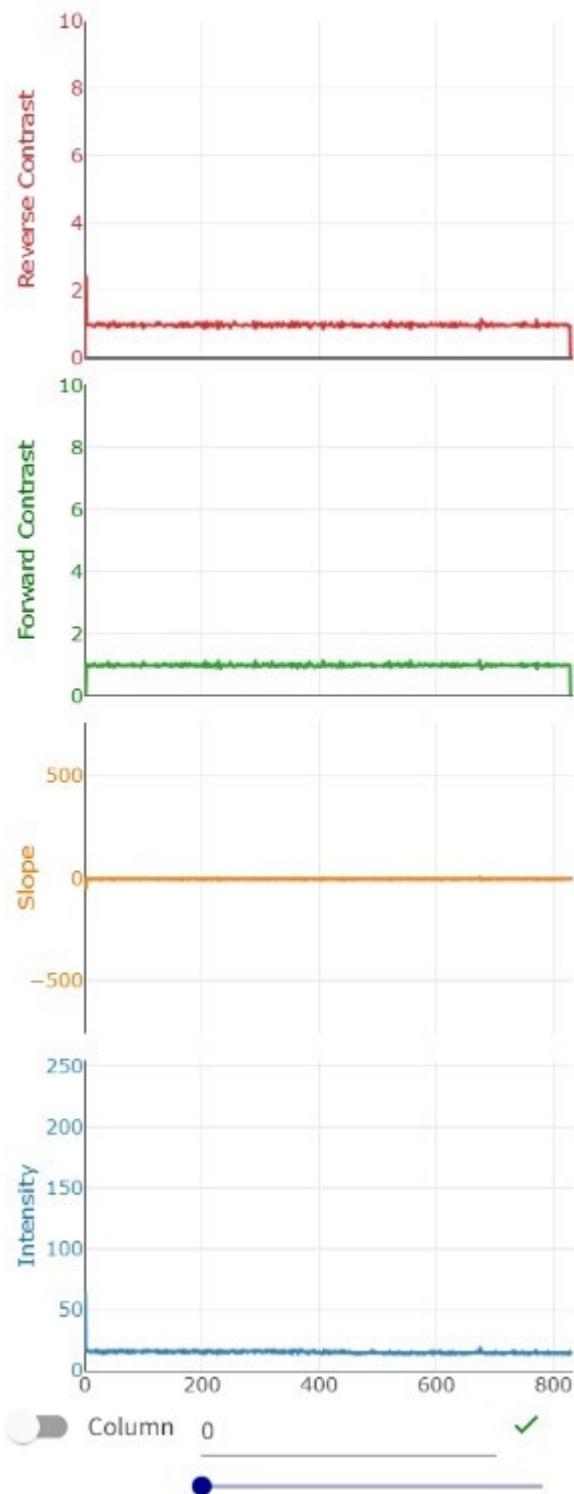


For more detailed analysis, additional settings and intensity graphs for individual rows or columns can be accessed by clicking the three lines on the far-right side of the dashboard. The slider button on the bottom of the window is used to cycle between row and column views, followed by the specific row or column you'd like to inspect. This helps BDF operators pinpoint specific areas of interest on a product.

Intensity Charts



Intensity Charts



XY Viewer: 3D Camera Calibration Check

To open the XY Viewer, select the "Start XY Viewer" button from the TDS2000 menu.

TDS2000_30140

 START RAW IMAGE VIEWER

 START XY VIEWER

 START CENTROID VIEWER

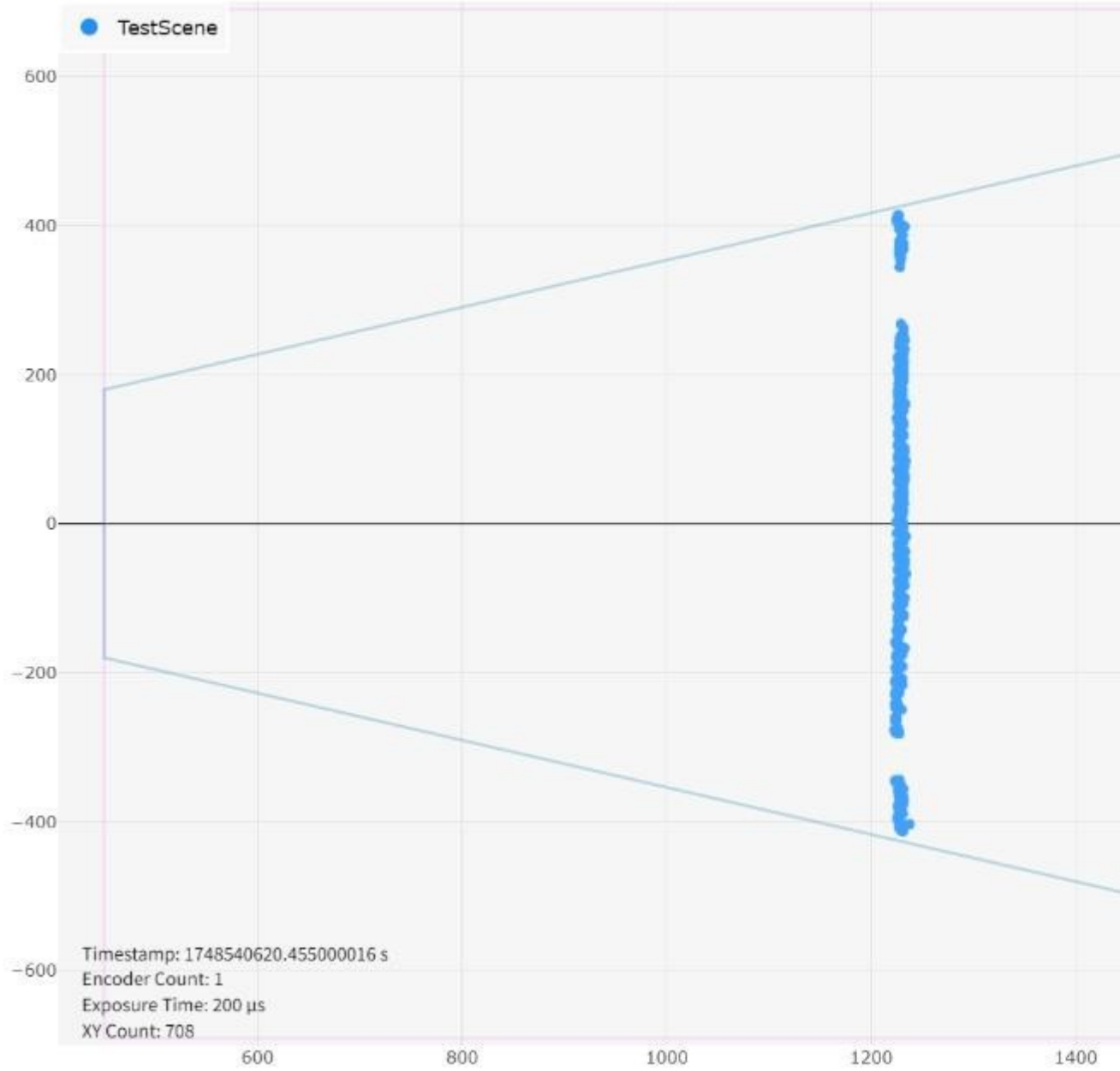
SET DEFAULT VALUES

> Category	Node Value
> Device Control	
> Acquisition Control	
> Image Format Control	
> Transport Layer Control	
> Application	
> Debug	

The XY Viewer is essential for verifying the mechanical calibration of your 3D camera. For accurate 3D measurements – crucial for consistent product dimensions and quality control in bakery automation – the line displayed in this viewer should be as close to vertical as possible after calibration.

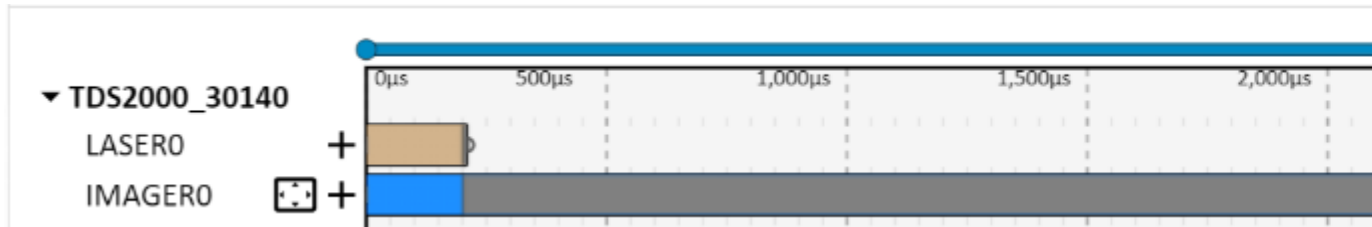
POINT CLOUD CAPTURE

XYs



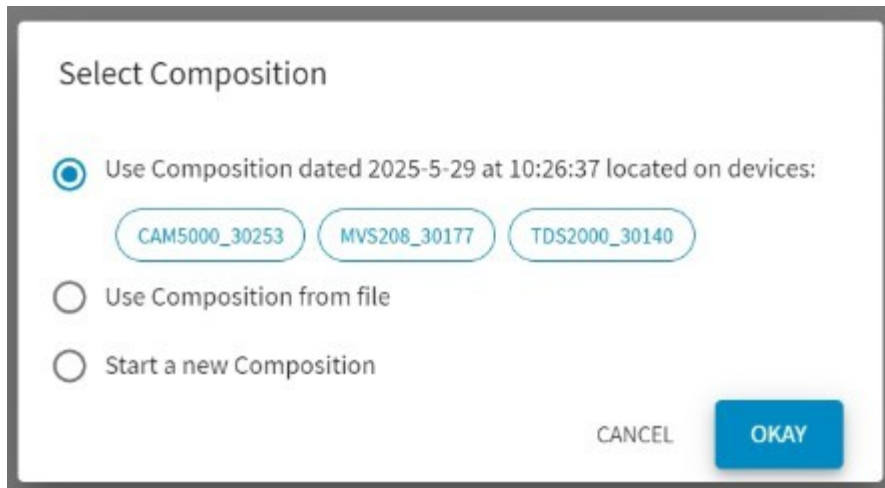
The Composer: Automating Your Bakery Inspection

The Composer is the advanced software application that allows BDF to configure and orchestrate the precise actions of your Hermary cameras. Think of it as the "brain" that tells your cameras exactly *when* and *how* to perform inspection tasks for your automated lines, offering unparalleled flexibility for custom bakery applications.



The diagram above illustrates a timing sequence for a camera and laser, showing how the laser is turned on and an image is captured simultaneously for 500 microseconds. This precision is vital for consistent quality control in a fast-paced bakery environment.

To begin creating a custom inspection sequence, launch the Composer through Amadeus Tools. Ensure all devices you plan to configure are powered on and connected to the same network.

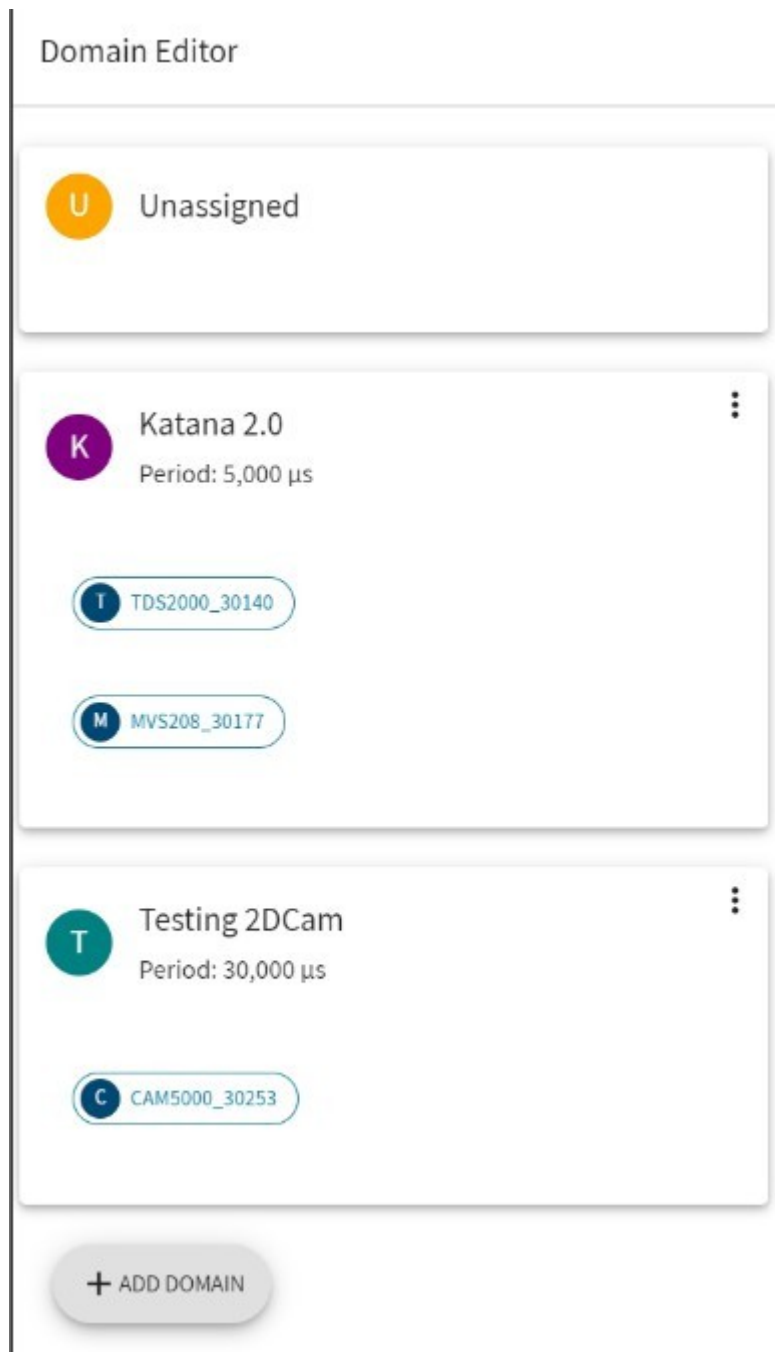


When first launching the Composer, you will have three options:

- Load an existing setup from a connected device.
- Load a setup from a file on your computer.
- Start a new setup from scratch.

Creating New Inspection Setups (Compositions)

Upon selecting "Start a new Composition," the main Composer window will open.



There are a few key elements to notice in this window:

- The section on the left is titled 'Domain Editor'.
- Your TDS2000_30020 camera is in a card under the 'Unassigned' Domain.
- The 'Add Domain' button.

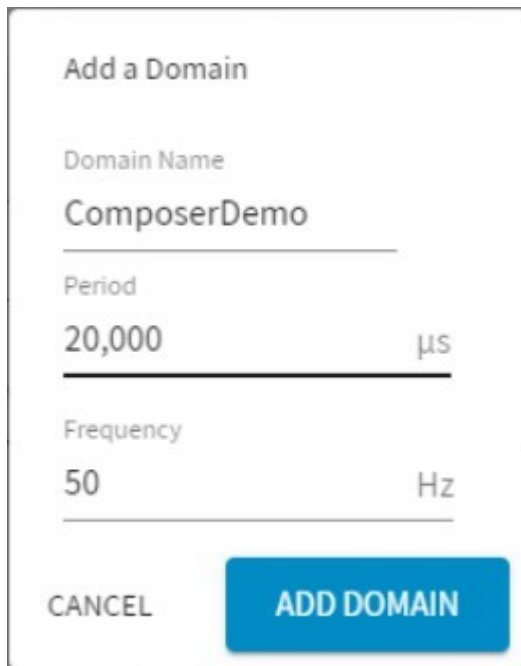
Understanding Domains

A 'Domain' is a group of devices (cameras) that will follow the same timing pattern for their tasks. To configure a device, it must be part of a domain. The next step in creating a Composition is to add a domain.

By clicking "Add Domain", a window will open. Here it is necessary to choose:

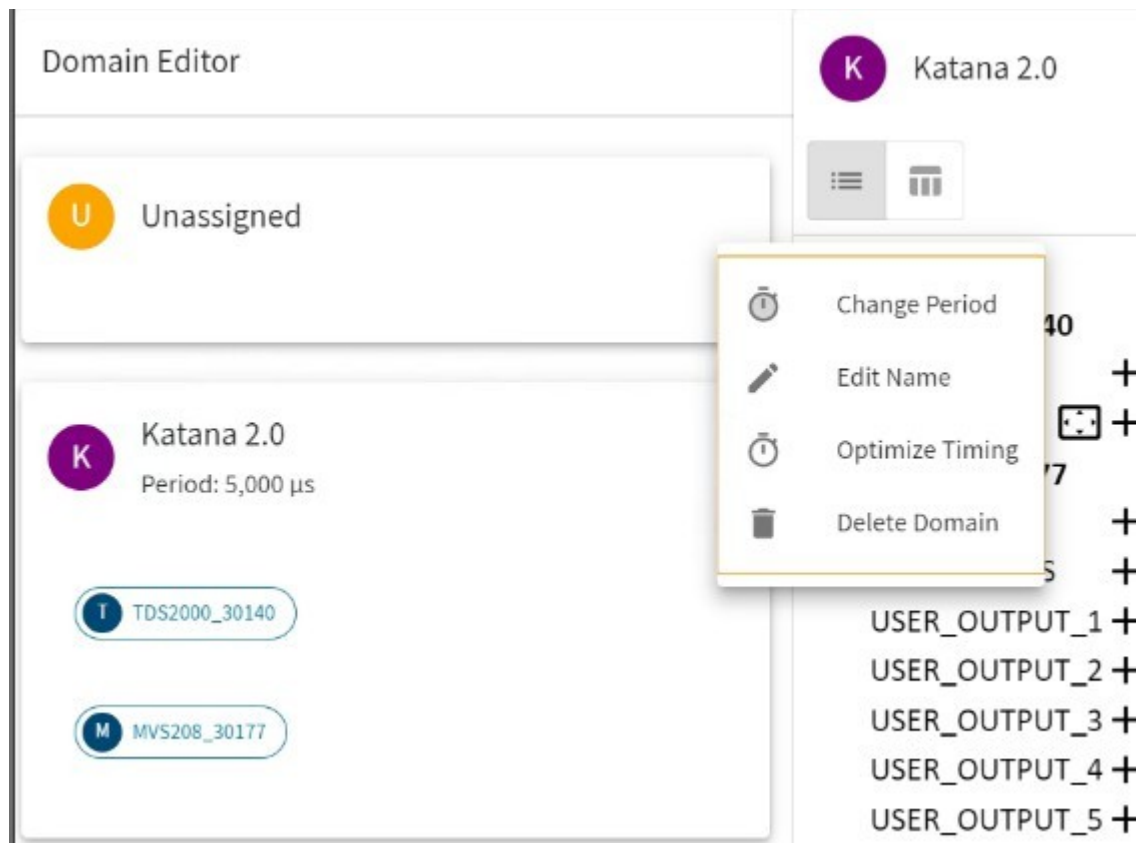
- A unique domain name, which will serve as an identifier (e.g., 'DoughInspectionLine').
- A period/frequency that the domain will follow (e.g., how often an inspection cycle repeats).

These two fields represent the same value in different units, and only one needs to be set. For example, a domain with a name of "ComposerDemo" might have a frequency of 50 Hz, meaning the period automatically changes to 20,000 μ s to reflect this.



The image shows a dialog box titled "Add a Domain". It contains two input fields. The first field is labeled "Domain Name" and has the text "ComposerDemo" entered. The second field is labeled "Period" and has the value "20,000" entered, with a unit selector showing " μ s". Below this, there is a "Frequency" field with the value "50" entered and a unit selector showing "Hz". At the bottom of the dialog, there are two buttons: "CANCEL" and "ADD DOMAIN".

You can edit domain-level settings such as the name, timing, and removal through the Domain settings menu, accessed through the dotted menu bar.



Assigning Cameras to a Domain

The main Composer window will update once a domain has been added. It is possible to have more than one domain, but most devices can only be assigned to one domain at a time. To add a device to a domain, drag and drop the device card (e.g., TDS2000_30020) from 'Unassigned' over to the target domain. The device will then be visible in the Composition Editor on the right.

AmadeusTools 4.6.7 - Composer

File View Help

 Amadeus TOOLS

Domain Editor

 Unassigned

 **Katana 2.0**
Period: 5,000 μ s

 TDS2000_30140

 MVS208_30177

 **Testing 2DCam**
Period: 30,000 μ s

 CAM5000_30253

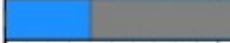
+ ADD DOMAIN

 **Katana 2.0**

▼ TDS2000_30140

LASERO + 

IMAGER0  + 

▼ MVS208_30177

ENCODER0 + 

USER_INPUTS +

USER_OUTPUT_1 +

USER_OUTPUT_2 +

USER_OUTPUT_3 +

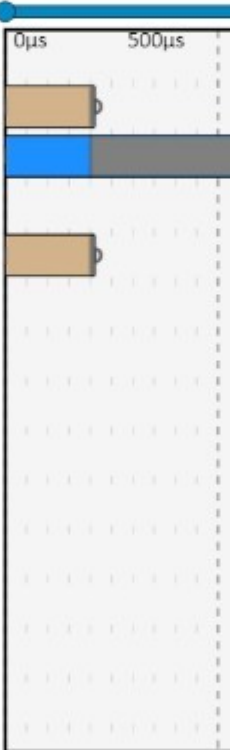
USER_OUTPUT_4 +

USER_OUTPUT_5 +

USER_OUTPUT_6 +

USER_OUTPUT_7 +

USER_OUTPUT_8 +



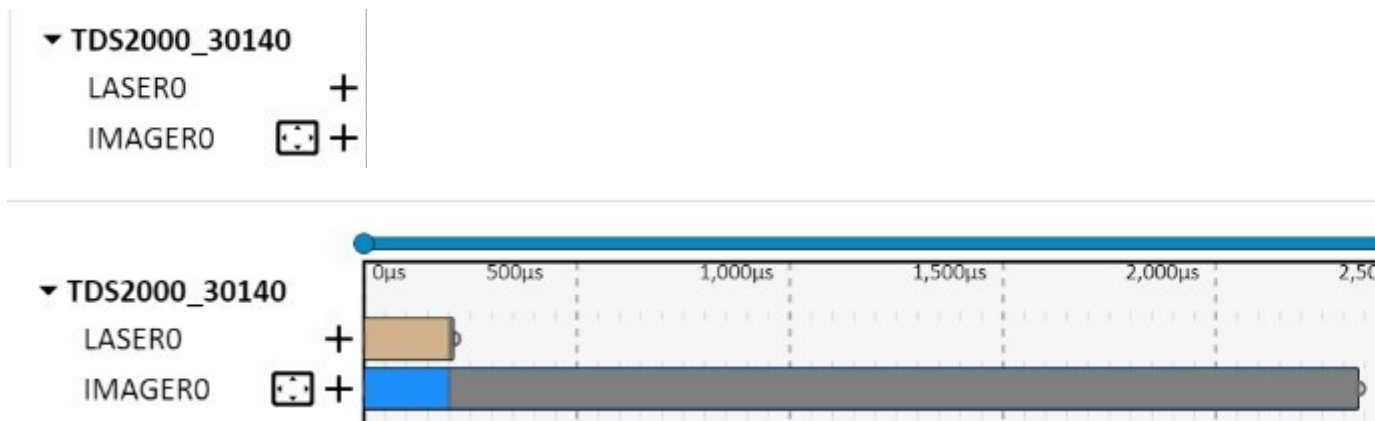
In the Composition Editor, you'll see:

- Your camera (e.g., TDS2000_30020).
- Its modules (like LASERO and IMAGER0), representing different functional components.

- A plus button next to each Module name to add a new Command for that Module.
- The main timeline, showing the operational sequence, with a scale along the top.

Modules are the functional parts within a device that provide a specific functionality that can be controlled by the Composer. For example, with the TDS2000, the LASER module controls when the laser turns on and off, and the IMAGER module controls when and how an image is captured.

Click the plus icon next to the module you'd like to add a command to. This defines a specific action (e.g., 'turn on laser' or 'take picture') to the Composition.



Clicking on a command in the Composition will select it. Once selected, the timing can be edited using the Command Editor Toolbar, or by clicking and dragging the command around the Composition. The command can be deleted by pressing the delete key after it has been selected.

There are certain restrictions in how commands can be timed, dependent on characteristics of the modules. If a command has a period where another command cannot be issued, this time is displayed as a greyed out area around the command, and the Composer will prevent other commands from being issued in this time.

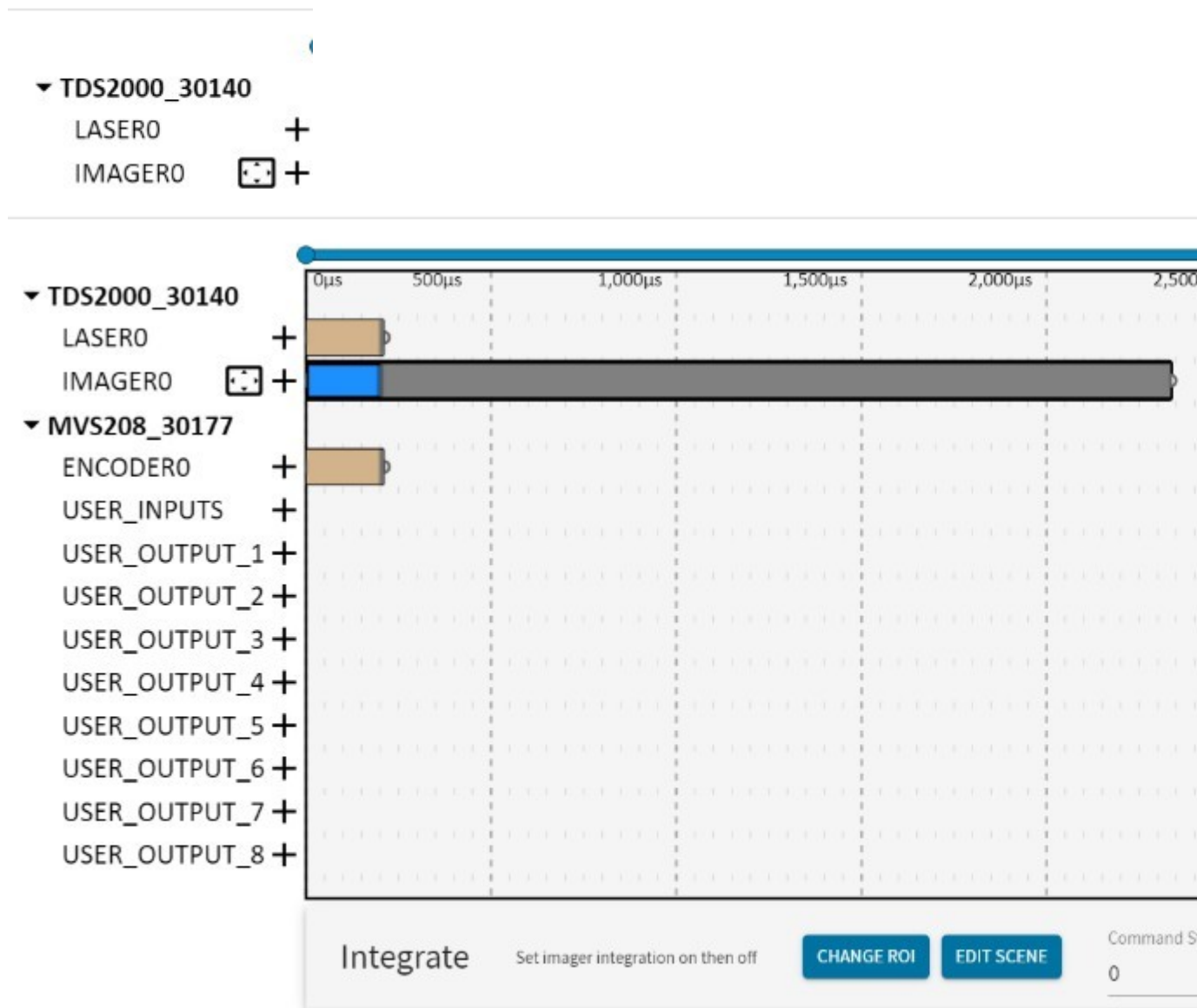
Changing Region of Interest (ROI)

To focus your camera on a specific area of your bakery product, you can change the Region of Interest (ROI). This helps save processing power and highlight crucial inspection zones, like the edges of a cookie or the surface of a cake.

You can change the region of interest by:

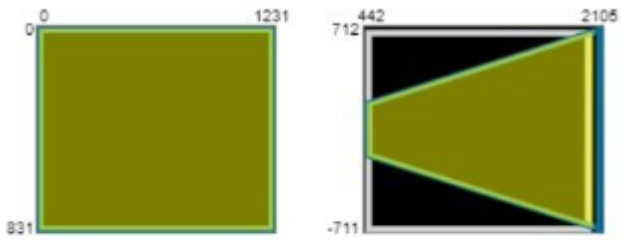
- Selecting the change region of interest shortcut icon beside the module.
- Selecting an integration command for the target module and selecting 'CHANGE ROI'.

The selected region of interest will apply to all integration commands for this imager module. The values entered may change after applying them; this is a function of how the imager works and is not avoidable.



The graphic below will update as you adjust the parameters. If you'd like to reset to the largest possible ROI, select "Max Extents".

Edit Pixel Region of Interest



Start Row	MinX
0	450
End Row	MaxX
831	2000
Start Column	MinY
0	-690
End Column	MaxY
1231	690

MAX EXTENTS CANCEL **OKAY**

Uploading Your Setup to Devices

Once your composition (inspection setup) is complete, upload it to your devices by clicking the "Upload Composition to Device(s)" button at the top of the screen and selecting the applicable devices. The pop-up will tell you when your changes have been successfully uploaded. Your BDF automation line is now ready to operate with these new inspection parameters!



Katana 2.0



▼ TDS2000_30140

LASERO



IMAGER0



▼ MVS208_30177

ENCODER0



USER_INPUTS



USER_OUTPUT_1



USER OUTPUT 2

